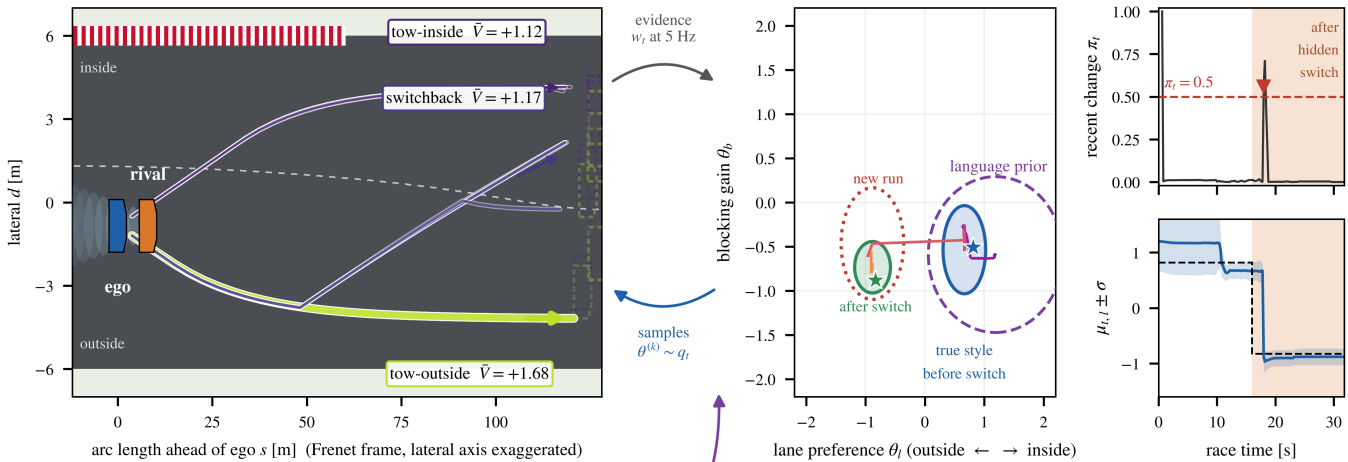

(a) Mental simulation of candidate manoeuvres
Futures imagined under 6 styles $\theta^{(k)} \sim q_0$. Dashed outlines are imagined rival replies.
tow-inside $\bar{V} = +1.12$
switchback $\bar{V} = +1.17$
tow-outside $\bar{V} = +1.68$
inside
outside
rival
ego
lateral d [m]
arc length ahead of ego s [m] (Frenet frame, lateral axis exaggerated)
evidence w_t at 5 Hz
samples $\theta^{(k)} \sim q_t$
(b) Belief over rival style
mean and $\pm 2\sigma$ of the run-length mixture $q_t(\theta)$
new run
language prior
after switch
true style
before switch
blocking gain θ_b
lane preference θ_l (outside ← → inside)
(c) Change points
recent change π_t
$\pi_t = 0.5$
after hidden switch
$\mu_{t,l} \pm \sigma$
race time [s]
prior
spotter radio (language input)
"car ahead covers the inside line"
evidence $(m_t, \lambda_t) = e_\psi(w_t)$
runs $q_t = \sum_r w_{t,r}\, \mathcal{N}(\mu_{t,r}, \Lambda_{t,r}^{-1})$
weights $w_{t,r+1} \propto w_{t-1,r}\,(1-h)\, p_r(m_t)$
residual $\hat{\pi} = v_{\max} \tanh(z_\nu + \omega^\top \chi),\ P_t = (1 + B_t^{-1})^{-1}$
decision $g_t^* = \arg\max_g \frac{1}{S} \sum_k \mathcal{M}(\hat{s}_t, g, \hat{\pi}(\cdot \mid \theta^{(k)}, \omega^{(k)}))$
radio prior $q_0 = \mathcal{N}(\mu_\ell, \Lambda_\ell^{-1}) = h_\phi(\text{msg})$